

Calcul Numeric

Problema 88.1 Sa se arate ca regula repetata a trapezului cu m subintervale, $T(m)$, este exacta pentru polinoamele trigonometrice al caror grad nu este multiplu de m . Ce rezultat da regula trapezului daca gradul este multiplu de m . (Indicatie: datorita liniaritatii este suficient sa se verifice exactitatea pe intervalul $[0, 2\pi]$ si functiile $f(x) = \cos kx$ si $f(x) = \sin kx$, sau chiar pentru $f(x) = e^{kix} = \cos kx + i \sin kx$).

Rezolvare:

$f:[a,b] \rightarrow \mathbb{R}$ integrabila pe $[a,b]$, $n \in \mathbb{N}$, $h = (b-a)/n$ si nodurile echidistante $x_k = a + kh$, $k = 0..n$.

Formula repetata a trapezului este:

$$\int_a^b f(x)dx = \frac{b-a}{2n} \left[f(a) + f(b) + 2 \sum_{k=1}^{n-1} f(x_k) \right] + R_1(f) \quad (1)$$

Definitia spune ca formula (1) este exacta pentru functia f daca $R_1(f) = 0$.

Polinomul trigonometric de grad p :

$$T_p(x) = a_0 + \sum_{i=1}^p (a_i \cos ix + b_i \sin ix).$$

$$\begin{aligned} \int_a^b T_p(x)dx &= \int_a^b a_0 dx + \int_a^b \sum_{i=1}^p (a_i \cos ix + b_i \sin ix) dx \\ &= \int_a^b a_0 dx + \sum_{i=1}^p \int_a^b (a_i \cos ix + b_i \sin ix) dx \\ &= \int_a^b a_0 dx + \sum_{i=1}^p a_i \int_a^b \cos kx dx + \sum_{i=1}^p b_i \int_a^b \sin kx dx \end{aligned}$$

Din enuntul problemei mi se da o indicatie: datorita liniaritatii este suficient sa se verifice exactitatea pe intervalul $[0, 2\pi]$ si functiile $f(x) = \cos kx$ si $f(x) = \sin kx$. Asadar, voi aplica formula repetata a trapezului pentru functiile $\cos kx$ si $\sin kx$.

$$x_k = a + kh \text{ unde } h = \frac{b-a}{n}, \Rightarrow x_k = 0 + k * \frac{2\pi}{n}$$

Inlocuim x_k in formule.

Formula repetata a trapezului pentru $\cos kx$ pe intervalul $[0, 2\pi]$:

$$\begin{aligned} \int_0^{2\pi} \cos kx \, dx &= \frac{2\pi}{2m} \left[\cos 2k\pi + \cos 0 + 2 \sum_{i=1}^{m-1} \cos \left(i * \frac{2\pi}{m} \right) \right] + R_1(\cos kx) \\ &= \frac{\pi}{m} \left[1 + 1 + 2 \sum_{i=1}^{m-1} \cos \left(i * \frac{2\pi}{m} \right) \right] + R_1(\cos kx) \\ &= \frac{\pi}{m} \left[2 + 2 \sum_{i=1}^{m-1} \cos \left(i * \frac{2\pi}{m} \right) \right] + R_1(\cos kx) \end{aligned}$$

Stim ca $\cos 0 = 1$ si $\cos 2k\pi = 1$, oricare ar fi $k \in \mathbb{N}$.

Formula repetata a trapezului pentru $\sin kx$ pe intervalul $[0, 2\pi]$:

$$\begin{aligned} \int_0^{2\pi} \sin kx \, dx &= \frac{2\pi}{2m} \left[\sin 2k\pi + \sin 0 + 2 \sum_{i=1}^{m-1} \sin \left(i * \frac{2\pi}{m} \right) \right] + R_1(\sin kx) \\ &= \frac{\pi}{m} \left[0 + 0 + 2 \sum_{i=1}^{m-1} \sin \left(i * \frac{2\pi}{m} \right) \right] + R_1(\sin kx) \\ &= \frac{\pi}{m} \left[2 \sum_{i=1}^{m-1} \sin \left(i * \frac{2\pi}{m} \right) \right] + R_1(\sin kx) \end{aligned}$$

Stim ca $\sin 0 = 0$ si $\sin 2k\pi = 0$, oricare ar fi $k \in \mathbb{N}$.

Voi lua pe rand cele doua formule. Intai voi lucra cu formula repetata a trapezului pentru $\cos kx$:

Voi incerca sa aflu valoarea integralei $\int_0^{2\pi} \cos kx$:

$$\int_0^{2\pi} \cos kx \, dx = \frac{1}{k} \sin kx \Big|_0^{2\pi} = \frac{1}{k} \sin 2\pi k - \frac{1}{k} \sin k0$$

Stim ca $\sin 0$ si $\sin 2k\pi = 0$, oricare ar fi $k \in \mathbb{N} \Rightarrow \int_0^{2\pi} \cos kx \, dx = 0$.

Pentru a rezolva suma $\sum_{i=1}^{m-1} \cos \left(i * \frac{2\pi}{m} \right)$ vom folosi formula trigonometrica a lui Langrage:

$$\sum_{n=1}^N \cos n\theta = -\frac{1}{2} + \frac{\sin(N + \frac{1}{2})\theta}{2 \sin \frac{1}{2}\theta}$$

Vom face inlocuiri: $N = m-1$, $\cos n\theta = \cos i \frac{2\pi}{m} \Rightarrow n=i$ si $\theta = \frac{2\pi}{m}$

$$\begin{aligned} \sum_{i=1}^{m-1} \cos i \frac{2\pi}{m} &= -\frac{1}{2} + \frac{\sin((m-1 + \frac{1}{2}) \frac{2\pi}{m})}{2 \sin \frac{1}{2} \frac{2\pi}{m}} = -\frac{1}{2} + \frac{\sin((m-1 + \frac{1}{2}) \frac{2\pi}{m})}{2 \sin \frac{\pi}{m}} \\ &= -\frac{1}{2} + \frac{\sin((\frac{2m-2}{2} + \frac{1}{2}) \frac{2\pi}{m})}{2 \sin \frac{\pi}{m}} = -\frac{1}{2} + \frac{\sin((\frac{2m-1}{2} * \frac{2\pi}{m})}{2 \sin \frac{\pi}{m}} \\ &= -\frac{1}{2} + \frac{\sin(\frac{(2m-1)\pi}{m})}{2 \sin \frac{\pi}{m}} = -\frac{1}{2} + \frac{\sin(\frac{2m\pi}{m} - \frac{\pi}{m})}{2 \sin \frac{\pi}{m}} = -\frac{1}{2} + \frac{\sin(2\pi - \frac{\pi}{m})}{2 \sin \frac{\pi}{m}} \end{aligned}$$

Pentru $\sin(2\pi - \frac{\pi}{m})$ vom aplica reducerea la primul cadran: IV – I: $\sin(2\pi - x) = -\sin x$

$$\sum_{i=1}^{m-1} \cos i \frac{2\pi}{m} = -\frac{1}{2} + \frac{\sin(2\pi - \frac{\pi}{m})}{2 \sin \frac{\pi}{m}} = -\frac{1}{2} + \frac{-\sin(\frac{\pi}{m})}{2 \sin \frac{\pi}{m}} = -\frac{1}{2} + \frac{-1}{2} = -1.$$

Vom inlocui rezultatul sumei in formula repetata a trapezului:

$$\int_0^{2\pi} \cos kx \, dx = \frac{\pi}{m} \left[2 + 2 \sum_{i=1}^{m-1} \cos \left(i * \frac{2\pi}{m} \right) \right] + R_1(\cos kx)$$

Si va rezulta:

$$\begin{aligned} \int_0^{2\pi} \cos kx \, dx &= \frac{\pi}{m} [2 + 2 * (-1)] + R_1(\cos kx) = \frac{\pi}{m} [2 - 2] + R_1(\cos kx) \\ &= \frac{\pi}{m} 0 + R_1(\cos kx) = 0 + R_1(\cos kx). \end{aligned}$$

De asemenea stim si valoarea integralei $\int_0^{2\pi} \cos kx \, dx = 0 \Rightarrow 0 = 0 + R_1(\cos kx) \Rightarrow$

$$R_1(\cos kx) = 0$$

Acum voi lucra cu formula repetata a trapezului pentru $\sin kx$:

Voi incerca sa aflu valoarea integralei $\int_0^{2\pi} \sin kx \, dx$:

$$\int_0^{2\pi} \sin kx \, dx = -\frac{1}{k} \cos kx \Big|_0^{2\pi} = -\frac{1}{k} \cos k0 + \frac{1}{k} \cos 2k\pi = -\frac{1}{k} + \frac{1}{k} = 0$$

Stim ca $\cos 0 = 1$ si $\cos 2k\pi = 1$, oricare ar fi $k \in \mathbb{N}$

$$\sum_{i=1}^{m-1} \sin \left(i * \frac{2\pi}{m} \right) = \sin \frac{2\pi}{m} + \sin \frac{4\pi}{m} + \sin \frac{6\pi}{m} + \dots + \sin \frac{2(m-1)\pi}{m}$$

Pentru a calcula suma, luam cate 2 termeni de la inceput si de la sfarsit si calculam suma acestora aplicand formula $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$.

Vom calcula suma a primului cu ultimul termen:

$$\begin{aligned} \sin \frac{2\pi}{m} + \sin \frac{2(m-1)\pi}{m} &= 2 \sin \frac{\frac{2\pi}{m} + \frac{2(m-1)\pi}{m}}{2} \cos \frac{\frac{2\pi}{m} - \frac{2(m-1)\pi}{m}}{2} \\ &= 2 \sin \frac{2\pi + 2m\pi - 2\pi}{2m} \cos \frac{\frac{2\pi - 2m\pi + 2\pi}{m}}{2} = 2 \sin \pi \cos \frac{2\pi - m\pi}{m} \\ &= 2 * 0 * \cos \frac{2\pi - m\pi}{m} = 0 \end{aligned}$$

Daca m-1 este par, suma va fi 0. Inlocuim in formula:

$$\int_0^{2\pi} \sin kx \, dx = \frac{\pi}{m} \left[2 \sum_{i=1}^{m-1} \sin \left(i * \frac{2\pi}{m} \right) \right] + R_1(\sin kx) = \frac{\pi}{m} 0 + R_1(\sin kx) = 0 + R_1(\sin kx)$$

De asemenea stim si valoarea integralei $\int_0^{2\pi} \sin kx \, dx = 0 \Rightarrow 0 = 0 + R_1(\sin kx) \Rightarrow$

$$R_1(\sin kx) = 0$$

Altfel, daca numarul termenilor din suma este impar, adica m-1 e impar suma va fi $\sin \frac{2\pi \frac{m-1}{2}}{m}$.

Daca luam 2 din fata sumei vom avea $2 \sin \frac{2\pi \frac{m-1}{2}}{m} = \sin \frac{2\pi \frac{m-1}{2}}{m} + \sin \frac{2\pi \frac{m-1}{2}}{m}$ Aplicam din nou

formula $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$ si vom

$$\text{obține } 2 \sin \frac{2\pi \frac{m-1}{2}}{m} = 2 \sin \frac{4\pi \frac{m-1}{2}}{2m} \cos \frac{2\pi \frac{m-1}{2} - 2\pi \frac{m-1}{2}}{2} = 2 \sin \frac{4\pi(m-1)}{4m} = 2 \sin \frac{\pi(m-1)}{m} = 2 \sin \frac{\pi}{m}.$$

Dupa ce vom face inlocuirile in formula vom avea

$$0 = \frac{\pi}{m} 2 \sin \frac{\pi}{m} + R_1(\sin kx)$$